The Hidden Variable in Embedded VLM Perception: Temporal Calibration for Real-Time Work-Zone State Estimation

Anonymous Authors
Paper under double-blind review

***Abstract*—Vision–language models (VLMs) are increasingly proposed to replace detector pipelines in driving perception, and such comparisons usually keep each system's published temporal smoothing constants. We show that these constants are a hidden variable that can decide the outcome. We run a fine-tuned INT4 2B VLM and a zero-shot 4B physical-AI reasoner on a Jetson AGX Orin and compare them with a YOLO+CLIP pipeline on 484K annotated frames of ROADWork driving, under a latency-honest protocol in which measured latency drives frame dropping. First, temporal constants do not transfer between models: inherited from the 2B, they leave the 4B model at macro-F1 0.290; recalibrated, it reaches 0.515. Second, the baseline needs the same treatment: recalibrated, it rises from 0.470 to 0.551 and no VLM-based system beats it, while a counted hidden-Markov filter on its score alone is equivalent to it on unseen drives. Third, constants do not transfer between devices either: on a DRIVE AGX Thor, cycle-counted constants degrade while the counted filter does not. The findings hold on drive-disjoint folds and at matched false-alarm rates and search budgets, and live concurrent execution reproduces the replay.**

I. Introduction

Work zones concentrate difficult driving-perception conditions: their appearance is regional and unstandardized, geometry changes daily, evidence such as cones, barrels, signs, and workers is small or occluded, and both misses and nuisance alerts are costly. They accounted for roughly 101,000 U.S. crashes and 899 fatalities in 2023 [1], yet production ADAS rarely model work zones explicitly and temporary speed limits are often absent from digital maps.

The strongest onboard approach combines a YOLO-style work-zone detector, CLIP verification [2], exponential smoothing, and hysteresis to estimate four traversal states [3], [4]. It is fast and effective after a zone is established, but reported IoU falls from 0.56/0.47 on OUTSIDE/INSIDE to 0.11/0.09 on APPROACHING/EXITING [3]. These boundaries are precisely where an advisory is most useful: an alert after the vehicle reaches the barrels has little anticipatory value.

Meanwhile, fine-tuned VLMs describe work-zone scenes well offline [5], and weight-only quantization plus embedded runtimes bring small VLMs to automotive hardware [6], [7]. Whether they should replace detector pipelines is usually judged by comparing systems as published, each with the smoothing and hysteresis constants its authors tuned. We build the strongest embedded VLM estimator we can, run it beside the detector pipeline on the same device, videos and protocol, and ask what such a comparison actually measures.

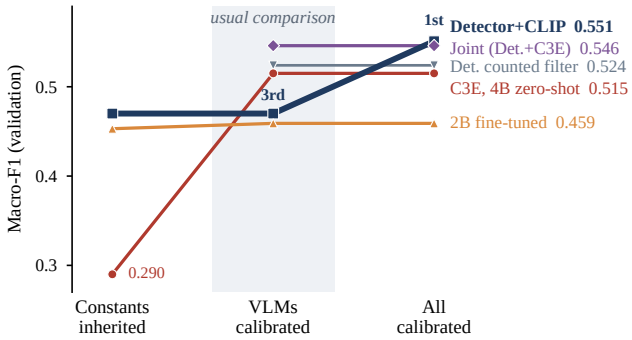


Fig. 1. The hidden variable. Validation macro-F1 (208 videos) under three comparison regimes. Left: every VLM runs the 2B's temporal constants and the detector pipeline its published ones; the zero-shot 4B reasoner (C3E) looks useless. Middle: the VLM systems are calibrated and the baseline is not, as in a typical comparison; C3E and the joint estimator beat the detector pipeline. Right: the baseline is calibrated by the same procedure and becomes the best system. Grey: the detector-only counted filter, which carries most of the joint estimator's lead.

The answer is: mostly the temporal constants (Fig. 1). Against the hand-tuned detector pipeline our VLM-based systems look competitive or better, but calibrating the baseline by the same procedure reverses the ranking. A team choosing a perception stack from the usual comparison would adopt a costlier VLM stack for no significant gain in macro-F1.

Our contributions are:

- **Evaluation:** a latency-honest protocol with per-model recalibration by evidence replay, confirmed by live concurrent execution on the Orin;
- **Findings:** temporal constants transfer neither between models nor between devices, and calibrating the baseline reverses the ranking, on unseen drives, at matched false alarms and with matched search budgets;
- **Systems:** a real-time evidence-first cascade for a quantized 2B VLM, an INT4 Cosmos3-Edge reasoner on the same device, and a counted hidden-Markov filter whose per-second parameters transfer across hardware, with an analysis of the VLMs' residual ego-relevance gap.

II. Related Work

a) Work-zone perception.: Early systems used environmental models with sensor fusion [8] or frame classification [9]. ROADWork [5] provides 4,375 videos and 9.6K annotated images from 18 U.S. cities and shows large gains from

in-domain fine-tuning (+32.2 AP, +36.7 SPICE). The closest system [3] deploys a ROADWork-trained YOLO detector with CLIP verification, EMA smoothing and hysteresis, and defines the traversal-state evaluation we adopt.

b) VLMs and world models for driving.: VLMs support planning [10], [11], description [12] and reasoning [13], and Cosmos world models plan at datacenter or automotive-SoC scale [14], [15]; we estimate one safety-relevant state under an embedded latency budget.

c) Cascades.: Detection [16] and LLM [17] cascades chain models of increasing cost and escalate on confidence; our stages share one model, and evidence semantics (transcribed text versus a binary answer) drive escalation.

d) Temporal filtering and fair comparison.: Our joint estimator is a discrete hidden Markov model [18] run as a Bayes forward filter [19]; we claim no novelty for it. That baselines deserve the proposed method's tuning budget is known [20]; we measure how much it changes conclusions for embedded temporal estimators.

III. System

A. Two embedded deployments

We deploy *two* models on the same Jetson AGX Orin 64 GB, which shares its SoC with the automotive DRIVE AGX Orin, adapted in two common ways: one by task fine-tuning, one by quantization and per-model calibration alone. Both run the same prompt channels, cascade structure, protocol and splits; only the model, its input size and its calibrated constants differ.

Fine-tuned 2B. A 2B-parameter VLM (Qwen3-VL architecture [21], initialized from Cosmos-Reason2-2B [22]), used for perception only, fine-tuned through a curriculum derived from ROADWork [5], whose imagery spans 18 U.S. cities under one signing standard (MUTCD [23]): work-zone VQA ($\sim$46K pairs), ego-state supervision, sign reading, a general-driving retention mix, and $\sim$10K work-zone-free BDD100K hard negatives. The last stage brought the existence gate from 95% to 0% "yes" on clean BDD100K scenes (on ROADWork OUTSIDE seconds, which include visible but irrelevant work, it is less selective, Table IV).

The language backbone is quantized to INT4 (W4A16 AWQ, group size 128) and the vision tower kept in FP16, both as on-device TensorRT engines [7] (1.38 + 0.83 GB). Decoding (10.4 ms/token) dominates cost, followed by vision encoding (44.6 ms at 720p, $\sim$20 ms at our 480px).

Quantized 4B world model. Cosmos3-Edge (C3E) [14] pairs an autoregressive reasoner with a trajectory-generating diffusion transformer; we run only the reasoner, and "world model" below refers to it. It is quantized to W4A16 INT4 (1.27 GB) with an FP16 visual tower (983 MB) on the same Orin; its frozen 26×46 patch grid fixes inputs at 736×416, and channel latencies are 162/182/488/556 ms (GATE/EGO/DESC/SIGN). It has seen no work-zone training data. Its engines required six patches to the embedded runtime, released upon acceptance.

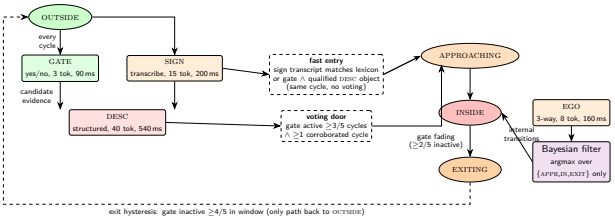


Fig. 2. Evidence-first cascade. In OUTSIDE, GATE and SIGN run every cycle and candidate evidence invokes DESC. Entry requires a sustained, corroborated vote or high-confidence same-cycle evidence. In active states, EGO drives a Bayesian filter; sustained gate clearance returns to OUTSIDE, while partial fading triggers EXITING. Constants are calibrated per model: $5/3/4/2$ for the 2B and $3/3/2/1$ for C3E, so the structure transfers but the numbers do not. Diagram values are the 2B's, with indicative latencies (measured values in Section V-C).

B. Prompt channels and their failure modes

We operate four fixed-prompt channels with fixed token budgets (the first generated token is an answer-start token).

a) Affirmation bias.: The GATE asks a neutral existence question, but binary prompts cannot be specialized by rephrasing: path-relevance variants all answered "yes" on OUTSIDE probes and distance prompts "close" regardless of scene. Binary outputs are therefore weak evidence requiring a temporal vote, never sole triggers. At $\sim$10 ms per output token, a three-token answer costs $\sim$85 ms and a description $\sim$500 ms, so evidence quality, not only confidence, gates smoothing.

b) Faithful transcription.: The SIGN channel returns verbatim text such as "SHOULDER WORK" and correctly reports absence, while a binary rephrasing misses signs that transcription catches; we match transcripts against a work-zone lexicon.

c) Structured corroboration.: DESC often verbalizes disqualifying context ("barriers on right *sidewalk*", "work vehicle *parked*"), so corroboration requires a work-zone object without such a qualifier. Prompts, budgets and lexicons are released upon acceptance.

C. Evidence-first cascade

The cascade (Fig. 2) allocates compute by state.

Outside (patrol). GATE and SIGN run every cycle (2B at 480px: p95 259 ms, 3.9 Hz), and either can invoke DESC. APPROACHING is entered by vote (gate active in K_{enter} of the last N cycles plus one corroboration) or by fast entry on a lexicon-matching sign or same-cycle gate+DESC agreement, which avoids a delay of up to N cycles ($\sim$1.3 s for the 2B) for evidence such as "ROAD WORK AHEAD."

Active states. EGO updates a Bayesian filter whose argmax is restricted to the active states, so that one "outside" reply right after a sign-triggered entry cannot erase it. Only sustained gate clearance returns to OUTSIDE; partial fading triggers EXITING, which EGO never produced. DESC refreshes every 1.5 s. The 2B's thresholds (N=5, K_{enter}=3, K_{exit}=4, K_{fade}=2) were fixed on calibration data; C3E gets its own (Section VI). With EGO disabled, as selected for C3E, APPROACHING becomes INSIDE on the transition prior alone, 13 cycles after entry.

D. Joint estimator over both evidence streams

Grafting the detector and the world model onto each other's temporal machinery does not work: the ported detector counts cycles (150-cycle maximum approach, 20 minimum frames outside) tuned for its native loop, and the cascade votes over a window tuned for the 2B's ∼260 ms cycle, while C3E answers only every ∼1.5 s. Either spine imports constants that are wrong for the combination.

The joint estimator therefore inherits no constant: it is a discrete hidden Markov model over the four states, run as a forward filter [18] with maximum-likelihood tables and no EMA, hysteresis threshold or cycle counter. Its observation combines the detector's frame score (six buckets), the world model's gate, sign-keyword and corroboration answers ordered by strength (six levels), and the age of that answer (three buckets): 108 cells. The emission is counted once per detector cycle on the calibration split with Laplace smoothing, jointly rather than factorized, because the interaction is the signal: the world model's APPROACHING is reliable when the detector corroborates it and mostly a false alarm when it does not (73% of those frames are OUTSIDE). The transition is a per-second rate matrix, counted as ground-truth transitions over dwell time and scaled by each step's dt, so it does not depend on the loop rate. The log-likelihood is scaled by dt/τ ($\tau = 0.5$ s): a world-model answer persists across ∼30 detector frames, and counting it once per frame drove an early version to 135 false alarms per hour. Bucket edges and τ were set on calibration data (Section VII).

The detector runs at its native rate and is never blocked; the world model is re-polled as soon as it answers, and its last answer is held with its age. We record raw evidence with measured clocks once per video and replay the filter on it, so ages and frame drops in every result are measured ones; Section VII confirms the replay against live concurrent execution.

IV. Evaluation Protocol

a) Benchmark.: We use the detector baseline's 520-video ROADWork traversal benchmark [3]: 4.31 h of 1920×1080 driving from 25 drives (30 fps except seven 120 fps videos), with 484K interval-labeled frames. A state-stratified 60/40 split assigns all design choices to calibration (312 videos, 2.59 h) and results to validation (208 videos, 192K frames, 1.73 h, 13 pure-negative). All 25 drives occur in both partitions, so Section VIII repeats every replay-fitted comparison with drive-disjoint folds. ROADWork samples its annotated images from the same drives as its videos, so the 2B's fine-tuning images, like the baseline detector's training images, may come from benchmark drives; C3E is zero-shot. Every calibration effect we report compares a system with itself, perception fixed, and is not inflated by this overlap.

b) Latency-honest camera simulation.: After each measured inference cycle of duration Δ on the target hardware, the replay advances by Δ and drops intervening frames as a live camera would, so slower systems see sparser frames and pay latency inside the accuracy result. Predictions are held between cycles and scored per frame.

c) Metrics.: We adopt the baseline's suite [3]: frame accuracy and macro-F1; per-state IoU; event precision/recall (an interval is correct if it overlaps one of the same state); entry offsets (Entry MAE over all states, IN entry med. for INSIDE); and false alarms per hour (predicted active intervals overlapping no annotated one), plus a lead metric for timing (Section V-A). Per-hour rates count frames at 30 fps (validation time overstated by 3%, for all systems alike). Intervals are paired 95% cluster-bootstrap intervals (10,000 replications) that resample drives.

TABLE I

VALIDATION, 208 VIDEOS, ONE PROTOCOL AND DEVICE (%; TIMING IN S). DET.+CLIP: HAND-TUNED [3] AND RECALIBRATED (SECTION VI); DET. *cnt.*: DETECTOR SCORE IN THE COUNTED FILTER; 2B: GREEDY, NO POST-FILTER (RECALIBRATED: 0.459); JOINT: DETECTOR+C3E IN THE COUNTED FILTER. LEAD: INSIDE ONSETS PRECEDED BY ≥2 S OF ACTIVE STATE. F1@20: AT ≤20 FALSE ALARMS/H (SECTION VIII).

Metric	Det.+CLIP hand	Det.+CLIP cal.	Det. cnt.	2B	C3E	Joint
Frame acc.	62.9%	64.8%	**68.2%**	62.0%	65.5%	66.1%
Macro F1	0.470	**0.551**	0.524	0.453	0.515	0.546
IoU OUT	0.617	0.627	**0.652**	0.581	0.615	0.641
IoU APPR	0.124	**0.296**	0.219	0.152	0.278	0.246
IoU IN	0.544	0.552	0.573	0.532	0.574	**0.578**
IoU EXIT	0.106	0.153	0.124	0.062	0.071	**0.160**
IN ev. recall	95.7	77.8	88.6	**97.3**	87.0	85.9
IN ev. prec.	79.9	**92.5**	75.9	56.8	71.5	66.8
False alarms/h	30.9	**20.2**	22.5	56.2	64.6	25.9
Entry MAE	2.52	2.31	2.44	2.52	**2.15**	2.22
IN entry med.	3.47	**2.10**	2.73	2.67	2.50	2.27
Lead ≥2 s	79.4	78.8	73.1	76.2	**86.2**	66.9
F1@20	0.410	0.529	0.520	0.409	0.487	**0.537**

V. The VLM Cascade Against the Published Baseline

A. Comparison with the deployed detector system

Table I places every system on the same videos, device and protocol. Against the detector pipeline as published, the greedy 2B ties on accuracy and macro-F1 (-0.018, $[-0.041, +0.003]$) and matches its INSIDE recall, but trails on precision (56.8% vs. 79.9%) and false alarms (56.2 vs. 30.9 per hour), most of them the ego-relevance errors of Section IX: a competitive detector-free result, not a replacement. A causal 1 s INSIDE debounce raises its precision to 76.0%. Its APPROACHING IoU is higher (0.152 vs. 0.124), as text and distant evidence appear before detector counts stabilize, a gap the recalibrated detector closes (Section VI).

B. Ablations and design validation

Table II toggles one component at a time from the sampled cascade (first row); five of seven change macro-F1 by at most 0.02. The sign channel must transcribe: an equal-cost binary question cuts accuracy from 62.5% to 52.5% and raises false alarms fivefold. Greedy decoding costs 0.5 accuracy

TABLE II
SINGLE-TOGGLE ABLATIONS OF THE 2B CASCADE (VALIDATION, 208 VIDEOS; –: NO INSIDE EVENT PREDICTED). LEAD: INSIDE ONSETS PRECEDED BY ≥2 S OF ACTIVE STATE.

Variant	Acc.	F1	IN prec.	FA/h	Lead
Cascade, sampled	62.5	0.464	54.1	60.1	79.4
greedy decoding	62.0	0.453	56.8	56.2	76.2
binary SIGN	52.5	0.407	46.1	305.8	86.2
no fast entry	63.1	0.462	54.3	34.3	69.4
no qualifier filter	60.3	0.447	53.9	69.7	81.2
no SIGN channel	63.2	0.456	53.4	47.8	82.5
unrestricted filter	61.4	0.445	53.1	69.1	73.8
base checkpoint	48.8	0.289	–	29.2	–

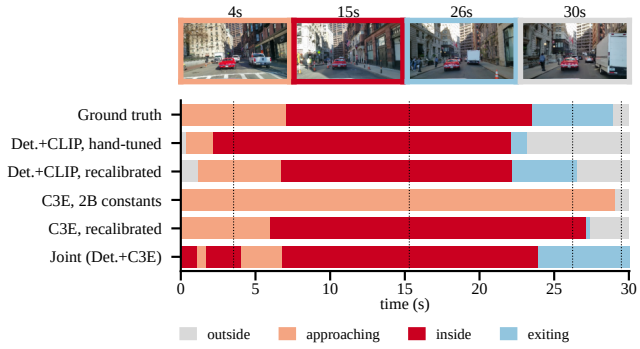


Fig. 3. Ground truth and five systems on one validation video, chosen to illustrate the effect; thumbnails mark the middle of each annotated state. With the 2B's constants, C3E never leaves APPROACHING; recalibrated, it tracks the zone. The hand-tuned detector pipeline enters INSIDE 5 s early and cuts EXITING short; recalibrated, it follows all four states.

points but improves precision and alarms, so we use it as the reproducible default. The qualifier filter adds 2.2 points and removes 9.6 alarms per hour. Fast entry buys anticipation, 10 more points of onsets warned ≥2 s ahead, for 26 more alarms per hour. The base checkpoint collapses under the 2B's thresholds (Section VI).

C. Operational cost

At 480px, p50/p95 latency is 83/90 ms (GATE), 136/149 (EGO), 155/169 (SIGN) and 468/508 ms (DESC); patrol and active cycles run at p95 259 ms (3.9 Hz) and 239 ms (4.2 Hz), the heaviest path at 768 ms. "Real time" means keeping pace with a live camera (259 ms is 7.8 m at 30 m/s), not an alert deadline. Above idle, the VLM draws 28.6 W and 3–26 J per cycle, the detector 22.4 W and 0.5 J at 24 ms.

VI. CALIBRATING EVERY SYSTEM

Section V ran each VLM with thresholds fitted to the 2B and the detector pipeline with its published constants. Here every system is recalibrated by the same replay procedure, and the ranking reverses (Fig. 1).

a) Temporal hyper-parameters do not transfer.: With the 2B's constants, the zero-shot C3E almost never reaches INSIDE (IoU 0.007, recall 3.8%) and looks useless. Recalibrating the same four integers and two switches lifts it to macro-F1 0.515 and INSIDE IoU 0.574, with model, prompts and videos unchanged (Fig. 3). The search covers $N \in \{3, 5, 7, 9\}$, $K_* \in 1..N$ and the EGO/SIGN switches (4,896 configurations, later extended to $N \in \{1, 2, 4\}$ without changing the optimum), selected on calibration macro-F1 over the 100 videos with recorded VLM evidence. The optimum (N=3, K=3/2/1) is shorter and less hysteretic and disables EGO (Section IX). Because each channel is recorded once and the state machine replayed, a sweep costs CPU-seconds instead of GPU-hours.

The same search moves the fine-tuned 2B by only +0.006 (0.453→0.459): it rescues a model whose constants were wrong rather than rewarding flexibility.

b) The baseline too.: The detector pipeline deserves the same treatment. We record every component on every cycle of all 520 videos (detector counts before and after CLIP verification, global CLIP and colour-context scores) and replay the complete pipeline; its hand-tuned configuration reproduces the live run (0.473 vs. 0.470). Searching its thresholds, EMA rate and duration limits on the 312 calibration videos (three grids of 4,800 configurations) raises validation macro-F1 from 0.470 to 0.551 ($+0.081$, $[+0.043, +0.111]$), with 92.5% INSIDE precision at 20.2 false alarms per hour: the strongest system we measure. On the detector score alone, without CLIP, the same search gives 0.435→0.530.

The procedure also isolates what fine-tuning buys: recalibrated on its own evidence, the base checkpoint reaches 0.383 on the search videos against 0.496 for the fine-tuned model, not the collapse it shows under the fine-tuned model's thresholds.

All choices are made on calibration data, VLM thresholds on the 100 videos with recorded VLM evidence and detector thresholds and counted tables on all 312, so the budgets favour the detector. Given the VLM's budget (one grid, the same 100 videos), the recalibrated pipeline still reaches 0.522, $+0.052$ over its hand-tuned version ($[+0.038, +0.070]$) and level with C3E.

c) No configuration dominates.: Calibrated, the world model and the detector pipeline tie on macro-F1 (-0.036, $[-0.080, +0.005]$) and fail differently: C3E leads INSIDE IoU and entry MAE, the detector pipeline precision and nuisance (20.2 vs. 64.6 false alarms/h). Section VII combines them.

VII. COMBINING THE DETECTOR AND THE WORLD MODEL

The joint estimator reaches 0.546 at 25.9 false alarms per hour: it beats the hand-tuned pipeline ($+0.076$, $[+0.038, +0.104]$) and the calibrated world model ($+0.031$, $[+0.006, +0.058]$), and ties the recalibrated pipeline (-0.005, $[-0.046, +0.033]$), trading precision for recall.

Its tables, counted on all 312 calibration videos (5-fold cross-validation 0.535, s.d. 0.019), are sparse but concentrated (20 of 108 cells carry 97% of the observations); 3 or 10 detector buckets give 0.538 and 0.545, and τ trades accuracy for nuisance (0.538 at 44.7/h for τ=0.25, 0.504 at 22.0/h for τ=1).

a) Concurrent execution.: The recorded runs behind every result had a 44 ms detector cycle (p50; p95 59 ms) and a world-model answer every 1.54 s. Running both models truly concurrently on the Orin, in wall-clock time from a decoded camera stream, on all 208 validation videos, raises the detector cycle from 24 ms alone to 53 ms (p95 58 ms, 19 Hz) with an answer every 1.44 s. Live, the joint estimator scores 0.548 against 0.546 in replay ($+0.002$, $[-0.011, +0.020]$; false alarms $+0.6$/h, $[-4.7, +5.6]$) and the detector-only filter 0.522 against 0.524: the replay is faithful under real GPU contention. The Orin draws 39.0 W median (46.9 W peak) with both models, 33.5 W with the detector alone and 11.1 W idle.

b) Where the gain comes from.: The detector carries the filter: without it macro-F1 drops by 0.086 and EXITING IoU from 0.160 to 0.017, while removing the world model costs 0.022 and lowers alarms (25.9 to 22.5 per hour). The detector score alone in the counted filter scores 0.524 at 22.5 false alarms per hour, tied with the recalibrated detector+CLIP pipeline on both (-0.026, $[-0.062, +0.009]$; $+2.2$/h, $[-8.1, +11.1]$) without CLIP or a threshold search, and 11.2 alarms per hour below the state machine recalibrated on the same score ($[-24.6, -2.1]$): the most efficient option here. The world model pays where the detector is weak: on the 10 validation onsets preceded by a transcribed work-zone sign, the joint estimator and C3E warn $\geq$2 s ahead on all ten, the recalibrated detector+CLIP pipeline on eight.

c) A learned alternative.: A strictly causal three-stage TCN [24] trained on the same evidence ties the best macro-F1 (0.550) but at 71.9 false alarms per hour.

VIII. Do the Conclusions Hold?

a) Unseen drives.: Splitting the 25 drives into five folds of about 104 videos, we refit the detector state machines (full three-grid search) and the counted filters on four folds and replay them on the fifth (standalone VLM evidence covers only 100 videos), so no test video shares a drive with its fitting data (Table III). The conclusions hold: recalibration raises the detector+CLIP pipeline by 0.054 macro-F1 ($[+0.026, +0.078]$) and the detector-score state machine by 0.089 ($[+0.058, +0.114]$), and both counted filters tie the recalibrated pipeline (-0.010, $[-0.036, +0.017]$; $+0.002$) and, with a ± 0.04 margin (half the recalibration gain), are equivalent to it by two one-sided tests (90% intervals within ± 0.032). On the validation videos, drive-disjoint fitting moves each system's macro-F1 by at most 0.010 from Table I.

b) Matched nuisance.: Systems operate between 18 and 65 false alarms per hour. With the same causal persistence filter applied to all (an active interval is shown only after lasting d seconds) until false alarms fall to 20/h (last column of Tables I and III), the calibrated systems lead: on unseen drives the recalibrated pipeline keeps 0.535, the joint estimator 0.523 and the counted detector filter 0.520, against 0.42 for the hand-tuned pipeline. Matching nuisance also reorders anticipation: at $\leq$30 false alarms per hour, C3E warns $\geq$2 s ahead of 58% of INSIDE onsets and the 2B of 54%, the counted detector filter of 73% and the recalibrated pipeline of 79%.

TABLE III

DRIVE-DISJOINT 5-FOLD REPLAY, ALL 520 VIDEOS: EVERY SYSTEM FITTED WITHOUT THE TEST DRIVES. F1@20: MACRO-F1 AT $\leq$20 FALSE ALARMS PER HOUR UNDER A SHARED PERSISTENCE FILTER.

System	F1	IoU IN	IN prec.	FA/h	F1@20
Det.+CLIP, hand-tuned	0.481	0.570	79.8	26.1	0.417
Det.+CLIP, recalibrated	0.535	0.560	**90.7**	**18.7**	**0.535**
Det. score, hand-tuned	0.434	0.546	73.3	47.0	0.410
Det. score, recalibrated	0.523	0.543	86.7	43.9	0.420
Det., counted filter	0.524	0.590	73.5	22.5	0.520
Det.+C3E, counted filter	0.537	0.591	70.8	28.7	0.523

TABLE IV

GATE "YES" RATE BY TRUE STATE, INSIDE-VS-OUTSIDE BALANCED ACCURACY OF THE BINARY ANSWER (BA), AND DESC CORROBORATION ON OUTSIDE; SAME 100 CALIBRATION VIDEOS, 1 HZ.

	GATE "yes" when truth is					
Checkpoint	OUT	APPR	IN	EXIT	BA	DESC
2B base	3.8%	53.1%	71.8%	9.8%	**0.84**	31.3%
2B fine-tuned	42.4%	85.3%	92.2%	51.2%	0.75	**10.7%**
C3E zero-shot	29.4%	93.3%	94.0%	34.4%	0.82	35.7%

c) Another platform.: We reran the hand-tuned detector+CLIP, both 2B decodings, C3E and the joint estimator on a DRIVE AGX Thor with Orin-fitted constants and tables. Model answers agree with the Orin's on 96–100% of identical frames, but every stage is faster (detector 16 vs. 24 ms; C3E's SIGN 157 vs. 556 ms), so cycle-counted constants span less time. The hand-tuned detector pipeline loses 0.031 macro-F1 ($[-0.037, -0.023]$) and C3E 0.042 ($[-0.055, -0.026]$), with 21 and 41 more false alarms per hour; the decode-bound 2B loses little macro-F1 (-0.007), though its alarms also rise by 21 per hour. The joint estimator, whose rates are per second and whose evidence is weighted by dt, keeps its macro-F1 without recalibration (0.549 vs. 0.546; equivalent within ± 0.04, 90% interval $[-0.015, +0.024]$). New hardware is the same hidden variable as a new model.

IX. Failure Analysis: the Ego-Relevance Gap

Nearly all residual false positives are correct descriptions of objects irrelevant to the vehicle's path (a parked work vehicle 60 m away, sidewalk work behind barriers, detour signs for another street); the qualifier filter catches only what the model verbalizes, and binary or distance prompts do not recover the judgment.

a) Response bias.: Recording the three checkpoints on the same videos separates how often a model says "yes" from how well it tracks the state (Table IV). The base 2B rarely says yes outside a zone and separates INSIDE from OUTSIDE best (balanced accuracy 0.84); fine-tuning moves the gate to 42.4% and 0.75; since ROADWork's OUTSIDE includes visible but irrelevant work, such a "yes" is often correct perception without ego-relevance. The fine-tuned model's advantage comes from its description, which corroborates on 10.7% of OUTSIDE seconds against 31.3% for the base model.

C3E, never fine-tuned, transcribes signs correctly yet labels most verified INSIDE frames "off-road/sidewalk" in ego-lane probes: perceiving objects and judging whether they claim the ego path are distinct capabilities, and no prompt we tried bridged them in either model. The joint estimator reaches 0.160 EXITING IoU only through the detector, whose score decays as objects recede (0.017 without it). ROADWork's 129K path annotations [5] could supply trajectory-grounded ego-relevance labels.

X. Limitations

Our claims are bounded in four ways. *(i) Data:* all footage is ROADWork daytime driving; drive-disjoint folds cover the temporal models, while the 2B's absolute accuracy may benefit from drive overlap with its fine-tuning images (Section IV). *(ii) Baseline:* comparisons use our port of the detector pipeline (0.470 on our split; 0.600 published on the original one). *(iii) Search:* configurations are the best found on finite grids; conclusions hold with matched budgets. *(iv) Scope:* replay is open-loop; we claim computational feasibility, not safety readiness.

XI. Conclusion

This study yields three insights for embedded perception. First, temporal constants belong to a system, not to a task: inherited across models they move INSIDE IoU by 0.567, and across devices they cost 0.03–0.04 macro-F1. Second, the baseline must be calibrated like the proposed system: doing so lifts the detector pipeline from 0.470 to 0.551 and removes the lead the VLM-based systems held over its published version. Third, parameters stated per second rather than per cycle transfer: a counted hidden-Markov filter on the detector score is equivalent to the calibrated baseline on unseen drives and keeps its accuracy on new hardware.

The practical implication is direct. Teams comparing an embedded VLM with a detector pipeline should recalibrate every system on the target hardware and compare at matched nuisance, and designers of temporal estimators should state their constants in seconds. The VLMs' remaining gap is ego-relevance, which trajectory-grounded supervision may close; night and adverse-weather evaluation with human labels is the next step. Upon acceptance we release the replay harness, prompts, metrics and baseline port.

References

[1] National Highway Traffic Safety Administration, "Work zone crashes and fatalities, 2023 (FARS and CRSS)," via National Work Zone Safety Information Clearinghouse, 2023.

[2] A. Radford *et al.*, "Learning transferable visual models from natural language supervision," in *Proc. ICML*, 2021, pp. 8748–8763.

[3] A. Martinez-Sanchez *et al.*, "ZONEWISE: Zone worksite inference with speed estimation," in *Proc. Int. Tech. Conf. Enhanced Safety of Vehicles (ESV)*, 2026.

[4] ——, "Vision-language work zone intelligence for safety-critical speed regulation of mixed-autonomy vehicles in dynamic environments," *arXiv preprint arXiv:2606.08860*, 2026.

[5] A. Ghosh *et al.*, "ROADWork: A dataset and benchmark for learning to recognize, observe, analyze and drive through work zones," in *Proc. ICCV*, 2025, pp. 6132–6142.

[6] J. Lin *et al.*, "AWQ: Activation-aware weight quantization for on-device LLM compression and acceleration," in *Proc. MLSys*, 2024.

[7] NVIDIA, "TensorRT Edge-LLM: High-performance LLM and VLM inference runtime for embedded platforms," https://github.com/NVIDIA/TensorRT-Edge-LLM, 2026.

[8] W. Shi and R. R. Rajkumar, "Work zone detection for autonomous vehicles," in *Proc. IEEE ITSC*, 2021, pp. 1585–1591.

[9] F. Abodo *et al.*, "Detecting work zones in SHRP 2 NDS videos using deep learning-based computer vision," in *Proc. IEEE ICMLA*, 2018, pp. 679–686.

[10] J.-J. Hwang *et al.*, "EMMA: End-to-end multimodal model for autonomous driving," *arXiv preprint arXiv:2410.23262*, 2024.

[11] X. Tian *et al.*, "DriveVLM: The convergence of autonomous driving and large vision-language models," in *Proc. CoRL*, 2024.

[12] A.-M. Marcu *et al.*, "LingoQA: Visual question answering for autonomous driving," in *Proc. ECCV*, 2024.

[13] NVIDIA, "Cosmos-Reason1: From physical common sense to embodied reasoning," *arXiv preprint arXiv:2503.15558*, 2025.

[14] ——, "Cosmos 3: Omnimodal world models for physical AI," *arXiv preprint arXiv:2606.02800*, 2026.

[15] ——, "Alpamayo-R1: Bridging reasoning and action prediction for generalizable autonomous driving in the long tail," *arXiv preprint arXiv:2511.00088*, 2025.

[16] P. Viola and M. Jones, "Rapid object detection using a boosted cascade of simple features," in *Proc. CVPR*, 2001.

[17] L. Chen, M. Zaharia, and J. Zou, "FrugalGPT: How to use large language models while reducing cost and improving performance," *arXiv preprint arXiv:2305.05176*, 2023.

[18] L. R. Rabiner, "A tutorial on hidden Markov models and selected applications in speech recognition," *Proceedings of the IEEE*, vol. 77, no. 2, pp. 257–286, 1989.

[19] S. Thrun, W. Burgard, and D. Fox, *Probabilistic Robotics*. MIT Press, 2005.

[20] G. Melis, C. Dyer, and P. Blunsom, "On the state of the art of evaluation in neural language models," in *Proc. ICLR*, 2018.

[21] Qwen Team, "Qwen3-VL: A versatile vision-language model series," https://github.com/QwenLM/Qwen3-VL, 2025.

[22] NVIDIA, "Cosmos-Reason2-2B model card," https://huggingface.co/nvidia/Cosmos-Reason2-2B, 2025.

[23] Federal Highway Administration, "Manual on uniform traffic control devices for streets and highways (MUTCD), part 6: Temporary traffic control," U.S. DOT, 2009.

[24] Y. A. Farha and J. Gall, "MS-TCN: Multi-stage temporal convolutional network for action segmentation," in *Proc. CVPR*, 2019, pp. 3575–3584.